

Failure log — current evidence, 2026-10-06

Author: Pankaj Yadav, Forward Deployed Software Engineer · github.com/Pankkaj64 · linkedin.com/in/pankkaj64

This is a development failure log, not the required full-room ground-truth evaluation.

1. **Counts existed but titles appeared blank.** In saved sweep `c7288b12-6942-43fe-9dff-44d909cde52c`, both models counted two books. The detailed output supplied subject summaries instead of visible locations and invented an author for *Extreme Ownership*. Those observations were rejected; the count survived as two anonymous candidates. Fixes: independent count/title status, OCR checks for authors as well as titles, explicit review reasons and rotated OCR. On the saved frame, the revised OCR found two lines (रावण, अमीश) in 1.102 s, versus one stored OCR line in the old record. That is a measured OCR-evidence improvement on one frame, not a demonstrated title-identification pass.
2. **Single-frame inference far exceeded a usable live cadence.** Local replay of `c7288b12-6942-43fe-9dff-44d909cde52c-1791272824025.jpg` took 430.235 s: primary count 85.519 s, independent count 5.676 s, lengthy detail generation, and a 180-second non-book timeout. The models agreed on two books; no title was confirmed and no reliable item list survived. The replay began before the final image/response-budget reductions. Counting now uses an at-most-800-pixel model image and 64-token output budget; original-resolution OCR/evidence is retained. Item prompts prohibit repetition and exact repeats are deduplicated. These latter performance changes have not been benchmarked on the full room. With two more weeks: profile Metal/offload, use an actual spine detector plus crop OCR, cache visual embeddings, and benchmark appropriate hardware/providers rather than promise a five-minute packet on this Mac.
3. **Moving shelves could erase pending evidence.** A single-slot queue replaced an earlier shelf with the newest frame while inference was busy. It now retains the newest pending view for each shelf, and optional continuous camera video preserves the pass. A regression test confirms Shelf 1's pending frame survives moving to Shelf 2; another verifies the final recorder chunk is included before upload. This fixes a reproducible data-flow loss, but does not provide automatic cross-view tracking. With two more weeks: implement geometric overlap/track association, spatial coverage and scale-aware spine segmentation.

Cost: local model API spend is zero; hardware/electricity and browser speech service costs are not measured. Optional market API access depends on the user's provider account. No paid source calls were made during these checks. New packets record stage duration, capture duration and time after capture stop. `data/diagnostics/workflow-replay.json` preserves the replay output without overwriting the original claim.

Submission gaps: no qualifying 60-book/two-unit/eight-item dataset, independent tape measurements, 15 checked prices, ten-book second-country demonstration or unedited audio demo. Their pass bars remain unmeasured. Some older frame files referenced by the existing packet are no longer present on disk; the app did not recreate or invent them.

Latest recording: inventory erasure and missed categories

Sweep `d2ad4b74-5627-45a7-9afa-5cec67c74d2e` had 29 frames but exported zero books and zero items. Each successful frame replaced all entries under `Shelf 1`; the last view contained no held book. Earlier frames and the video survived.

Repairs: accumulate one-to-one observations; preserve reviewed facts, offers and measurement evidence; keep historical boxes; retain every queued sample; include room detections in frame history and the UI. YOLO26s misclassified Extreme Ownership as a phone. A constrained crop reader recovered that title, 8 Rules of Love, and रावण. One full model replay took 88.87 s and retained all three through the final empty view. Titles are proposals, not automatically confirmed identities.

Rotated OCR results previously mixed conflicting readings, and one invalid role selection discarded a whole batch. The pipeline now selects one coherent OCR orientation and handles books separately. Author/publisher text may still be unreadable; leaving it blank is intentional.

The added YOLOE room detector covers more required categories but produces TV/ mirror mistakes, false positives and duplicate partial-object candidates. Scene alignment reduces duplicate tracks; it does not establish a physical count. The experimental contact-sheet room reader timed out or shifted categories (cup to chair), so it is disabled and documented in `model_choices.env`. No $\geq 80\%$ room-item accuracy claim is made. More suitable local hardware/models or a trained room/ spine dataset, plus independent ground truth, are still needed for acceptance.

7 October 2026: pricing, export contract and portability fixes

4. **The exported `claim_packet.json` dropped required fields.** The download only contained a `materials` array of verified names. Root cause: the export filter written to keep unverified guesses out of the report was also used for the deliverable. Impact: an adjuster could not trace prices, totals or the review queue. Fix: `claim/contract.py` emits the brief's structure (required keys first; unknowns `null/""`; totals recomputed in code from the exported lines) and the report is rendered from it. Verified against the brief's key order, blanks, totals and sources.
5. **Prices existed only when an operator typed them.** Research returned eBay candidates but nothing applied them, and there was no FX path, so foreign quotes were dropped silently. Fix: provider quotes carry a title/author match score; valuation applies them deterministically (operator > local provider > converted provider), records every rejection reason in `price_details`, converts with dated ECB/ER-API evidence, and re-prices ten books for a second country. Verified with mocked providers; real quotes still need eBay credentials.
6. **OCR only worked on macOS.** Spine text used an Apple Vision helper compiled with `swiftc`; on any other machine it fell back to an untested path. Fix: `perception/ocr.py` with PaddleOCR (default) or EasyOCR behind one function, selectable with `OCR_ENGINE`. Not yet measured: OCR recall on the 60-book shelf with the new engine.

7 October 2026: first real run on the development laptop

7. **Nothing was detected in a live sweep (a held water bottle, three frames).** Every frame carried `Detection failed: ModuleNotFoundError: No module named 'ultralytics': the Python that served the API had none of the perception packages, no YOLO weights had been downloaded and Ollama was not running.` The guidance line did say "I could not analyse that view", but nothing else in the UI made the failure obvious. Fix: install `requirements.txt` into the interpreter that runs `uvicorn`, run `tools/setup_detector.py`, start Ollama; `GET /api/health` now reports the configured models. With the stack in place the same kind of frame yields television, chairs, vases, lamp, rug and framed painting on a living-room test image in under three seconds.
8. **PaddleOCR could not be imported on a machine that also had TensorFlow + Keras 3.** `paddleocr imports transformers`, which tried to initialise its TensorFlow backend and raised. Fix: `logic/ocr.py` sets `USE_TF=0` before the import. Measured afterwards: PaddleOCR read 5/6 synthetic spines (the miss was text cut off at the strip edge) and real spine fragments ("CATALONIA", "COL-LECCIONS") where EasyOCR read nothing and misread "Dune" as "auna". PaddleOCR stays the default; 0.3-0.4 s per spine once loaded.
9. **Qwen2.5-VL 3B cannot load on an 8 GB Mac.** Ollama's Metal backend failed to allocate a 6.6 GB buffer; with a 1 k context it loaded but took 57-160 s per crop, far beyond the 30 s reader timeout, so no title was ever proposed. Fix: Gemma 3 4B is the reader (5/5 synthetic titles exact once warm, 8-10 s per crop), the reader timeout allows a cold load, models are kept resident for 30 minutes and warmed at startup. Moondream was tried as a lighter reader: it loaded slowly (four timeouts) and then read 2/2 correctly in 4-6 s.
10. **No book could ever be identified on a full shelf.** The identity gate required the whole frame's book count from YOLO26s and YOLO11s to be equal; on real shelves the two COCO models gave 48 vs 3 and 9 vs 17 boxes. Fix: the check is per book (a second detector drew the same box, `IoU >= 0.5`); the frame-wide count remains a review note. Also, the YOLOE room detector's "book" boxes were being discarded: on a shop wall it found 18 spines where the COCO detectors found 1-2, and 51 vs 9-17 on a library shelf. They now join the candidates with the detector name recorded as a witness.
11. **The blind crop check rejected every real book.** Gemma 3 4B, asked for the category of a single spine in a stack, answered "table" or marked the crop as not a clear single object, so no book could pass the identity gate. Qwen3-VL 2B agreed on 5/6 real crops with the cover, spine and title text named, but only after `/no_think` was appended to the prompt: its Ollama `think` flag is ignored and the JSON never arrived. Fix: Qwen3-VL 2B is the verifier, Gemma 3 4B the reader, so the two opinions still come from different models.
12. **Google Books returned nothing even with a working key.** The provider searched with `intitle:"..." inauthor:"..."`; on 8 October 2026 those operators (and `isbn:`) return zero volumes for well-known titles, while a plain-text query returns hundreds. Fix: plain query, 20 results, the existing title/author match score does the filtering. Measured after: 6 candidates for

"Zero to One" in GB, 3 in AE. Remaining limit: every priced volume Google returned was a Google Play ebook, which the valuer rejects as a physical replacement cost unless `PRICING_ALLOW_EBOOK_PROXY=true`; physical prices need the eBay source.

13. **A fixed OCR budget skipped most spines on a dense shelf.** With a 10 s per-frame OCR budget, 40–52 of 53 worn spines were never read ("OCR frame budget reached"). Fix: the budget scales with the detected spine count (10 s floor, 1.5 s per spine, 60 s cap). Measured after, same photo: 48 of 53 spines read (was 13), 3 identified (was 1). Cost: the title reader then runs on 48 crops at 8–12 s each, so the frame took 13 minutes; the reader needs the same budget-and-defer treatment as the crop check.
14. **An author's name was accepted as a title.** On the worn shelf, OCR read Northanger , JANE AUSTEN , Abbey ; the reader proposed "JANE AUSTEN" with an empty author, and every gate rule passed (exact OCR text, confidence 0.985, two detectors). The gate could not tell a writer's credit line from a title. Fix, three layers: the reader prompt now says a person's name belongs in author , and a title equal to the chosen author is dropped; the live gate refuses a title that equals the author line; and at finish the Open Library authors search is consulted first, so a "title" that is an author with 20+ works is demoted to author with a review finding. Measured after, same packet: "JANE AUSTEN" demoted (2,225 works), identified count 3 -> 2, the name kept as the line's author. Remaining limit: the live gate alone still passes this case when the reader leaves author empty; the catalogue check needs the network at finish time.
15. **Stop sweep left the sweep hanging with "Wait for frame processing before finishing".** Frames go out every 2 s and take 20-60 s each, so Stop almost always lands while one is in flight. The frontend turned the camera off and called finish at once; the backend's 409 (correct: merging a frame during finish would corrupt the packet) was shown and never retried, so the sweep stayed active for ever. Fix: Stop waits for the in-flight frame, retries finish on that 409, and a status strip shows scanning / processing / finishing. Verified in the browser with a synthetic camera feed: Stop pressed at 30 s into a 50 s frame, finish returned 200 after the frame, no 409 in the network log.
16. **A tissue box counted as a book and the blind check crashed on it.** The detector called the box a book at 41 %, the second detector agreed on the box, OCR read the printed marketing text, and Gemma proposed the manufacturer as the author. The Qwen blind check, the one model that should have said "not a book", answered a category outside its list and the code raised a ValueError, recorded as "unavailable". Fix: an unlisted answer such as "tissue box" is kept as raw_category and becomes a conflict with the reason "Blind check saw a tissue box, not a book". The gate had already kept the line unidentified; the box still counts as one detected book, which is a known limit of the COCO "book" class.
17. **Three of 56 books identified on a worn shelf, and the OCR threshold was not the cause.** Measured per spine with all other rules held: at 0.95, 0.9, 0.8, 0.7 and 0.5 the same three spines pass; no spine was blocked only by confidence, and the lines under 0.9 are the garbled ones ("DDEI", "TTIME", "CLIKIN"). The real reasons, overlapping: box touching the frame edge 29, no second detector 30, reader gave no title 24, title under 8 characters 16, proposal not in OCR 5. The one wrong identification, "MASTERPIEG", came from a line at 0.99. Fixes, in

order: edge/closer guidance; evidence pooled across frames with scene-aligned book tracks; OCR upscale, contrast and 180°; the blind check's transcription as a second reading with a `two_models` path; misread refusal when the two readings differ by a letter; Open Library confirmation or misread flag at finish.

18. **Tiled detection and the first pooling attempt inflated the count to 129.** Running the three detectors on overlapping tiles found 16 more boxes on the whole-shelf photo, nearly all edge-cut, and YOLO11s found 2 boxes at tile scale, so no second witnesses were gained. With the shelf sent as three overlapping views and tracks matched by appearance only, 158 frame boxes became 129 lines (expected about 53). Fixes: tiles behind `DETECTOR_TILES`, off by default; book tracks use the scene transform between views (projected overlap of at least 0.45), which cut the first two views to 60 lines with 26 seen in both. The third view never arrived because the backend was restarted mid-run; the complete measurement is still to be taken. Remaining truth about this shelf: even across two views 34 of 60 spines never appear whole, because the source photo is itself a tight crop.
19. **Five Hindi books, none read.** OCR ran with `OCR_LANG=en`, so Devanagari spines came out as Latin noise ("अमीश" as "3TH2T"); the blind check read them correctly ("अमीश श्रीमद्भगवद्गीता") but the gate needs OCR's own reading. Two further faults surfaced: the shared word matcher dropped Hindi vowel signs, so even a correct Hindi reading would have failed every title check, and the 10 s OCR floor was spent on four English rotations before any Hindi pass could start. Fixes: a Hindi (Devanagari) OCR pass tried right after a weak upright English read; Hindi-safe `normalized`; budget per OCR language with a 30 s floor. Measured on the saved frame: the Gita spine now reads "श्रीमद्भगवद्गीता" (0.93) with author "आर.आर. वर्मा", and "राम" / "अमीश" for Ram by Amish. Still unidentified, correctly: every box touches the frame edge, "राम" is under the 8-character minimum, and OCR dropped one letter of the Gita title, which the misread rule refuses. On the 8 GB laptop the two languages took 107 s for the frame inside the app (2–8 s per crop when tested alone), so the last two books ran out of OCR time.
20. **A 17-frame English sweep identified nothing because Ollama had stopped.** OCR read the covers well ("RULES / OF LOVE / JAY SHETTY", "EXTREME / OWNERSHIP") but every frame logged "Local title reader unavailable (ConnectError)" and every blind check was "unavailable", so no title was proposed. Restarting Ollama brought all three models back. Remaining gap: the status strip does not say that the model server is down; that belongs to point 1 of the next-week plan (failures shown in red).